

A New Single Stage AC-DC Converter for Low Voltage Electromagnetic Energy Harvesting

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Abstract -- In this paper, a direct AC-DC converter for electromagnetic microgenerators is presented. In recent years, researchers have developed a number of microgenerator configurations that are capable of producing an AC voltage of a few hundred millivolts. However, little work has been reported on conversion of such low voltages to a steady DC voltage required by most devices. The authors present a new converter topology that can be used for such low voltage, low power energy harvesting. It utilizes the bidirectional current capability of MOSFETs to avoid the use of front-end bridge rectifier. The converter offers a resistive load to the microgenerator making it suitable for optimum energy harvesting. Simulation and experimental results are depicted to verify the operation of the converter.

Index Terms--AC-DC power conversion, energy harvesting, low power, low voltage, Buck-boost conversion

I. INTRODUCTION

Vibration-based energy harvesting [1-3] has been a relevant research topic in recent years. In particular, Electromagnetic microgenerators [4, 5] have been quite popular because of their high energy density (Fig. 1). They are best suited for applications requiring a few milliwatts of power. Such microgenerators work on the principle of Faraday's law of electromagnetism. The mechanical vibration is utilized for the movement of a permanent magnet. The moving magnet induces emf in a stationary coil. The amount of harvested energy can be controlled by changing the load resistance connected to the coil. The major challenges in electromagnetic energy harvesting are dedicated design and power processing.

Most of the electronic devices typically require steady 2-3.3V DC for their operation. However, induced emf for microgenerators is quite low. For most of the reported topologies, the induced emf is around a few hundred millivolts. Therefore, conventional AC-DC converters involving a full bridge rectifier cannot be utilized [6-8]. A converter is required that can convert AC in a range of few hundred millivolts to a nominal DC voltage. However, the topology cannot involve a traditional rectifier-based approach due to the obvious limitation of overcoming diode voltage drops ($V_f > 0.2V$).

Some researchers have proposed novel topologies for such single stage AC-DC boost conversion. Mitcheson et. al. [9] reported a dual polarity boost converter which utilizes two

boost converters for each half-cycle of the input voltage. The converter requires two inductors and input polarity sensing for its operation. Two more direct AC-DC converters were also reported in [10, 11]. The researchers aimed to utilize a single inductor for boost operation in both the half-cycles.

The current work presents a topology based on a buck-boost configuration. The buck-boost configuration has obvious advantages over the few boost converters based topologies presented in literature – (a) the main advantage is that the power extracted from the input only depends on the duty cycle of the converter, and not on the load characteristics. Such a scheme offers a resistive load to the microgenerator and is very suitable for optimum energy harvesting; (b) it allows a voltage feedback scheme for the converter simplifying the control structure; (c) the inrush current can be controlled.

In this work, a direct AC-DC converter using a single inductor is proposed (Fig. 2). The converter utilizes a bidirectional switch to perform the voltage conversion. The diodes, D1 and D2 utilized in this converter are on the output side. The average current through them is low, and hence, the losses when compared to a front-end bridge rectifier are significantly less. The converter is operated in discontinuous conduction mode (DCM) to reduce switching

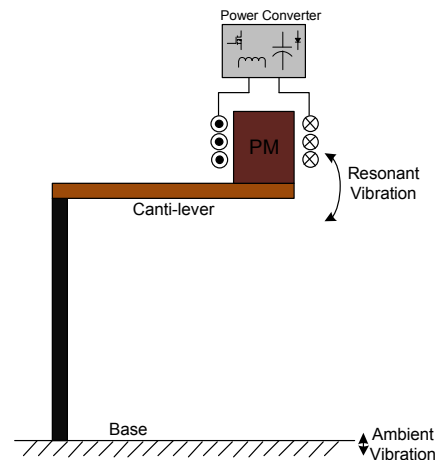


Fig. 1. Microgenerator-Converter System

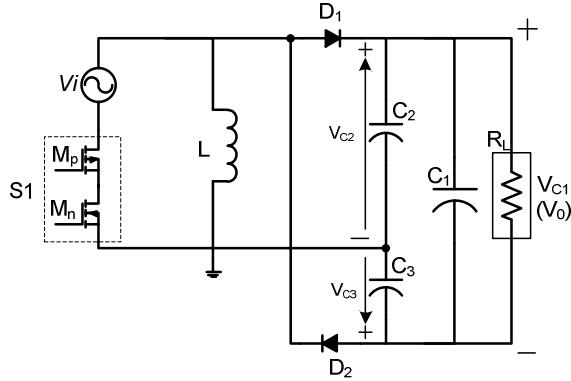


Fig. 2. Proposed AC-DC converter topology

losses and enable simple feedback control. The rest of the work is distributed as follows. The detailed operation and analysis of the converter is discussed in section II. The simulation results for the converter with low input voltage are presented in Section III. Section IV presents the experimental results for the proposed converter. Finally the paper is concluded.

II. CONVERTER OPERATION

In this section, operation and analysis of the proposed topology is presented.

A. Topology and Operation

The proposed converter is shown in Fig. 2. It consists of a single inductor for boost operation in both the half-cycles. The bidirectional switch comprises of n-MOSFET, M_n and p-MOSFET, M_p (Fig. 3). This switch is referred as S1

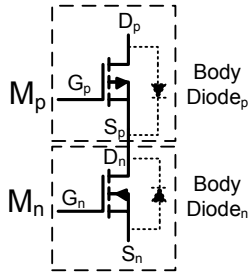


Fig. 3. Bidirectional Switch

throughout the paper. The source of MOSFET M_p is connected to the drain of MOSFET M_n . As shown in Fig. 3, such an arrangement causes the MOSFET body diodes to have opposite orientation. The MOSFETs are turned on together to act as a single switch, M_n with a positive gate pulse while M_p with a negative gate pulse. When the switch is on, inductor current builds up. On turning off the switch, the body diodes block the circulating current and the energy is transferred to the output side. Therefore, the proposed switch S1 has both bidirectional current carrying and blocking capability.

The converter has four main operating modes as shown in Fig. 4. Mode I and II are for the positive half cycle while modes III and IV correspond to the negative half cycle.

1) Positive half-cycle:

Mode I: Both the MOSFETs, M_p and M_n are turned on. The input voltage is applied to the inductor. The inductor current builds up linearly in the positive direction from zero.

Mode II: The switch S1 is now turned off. The circulating current cannot flow through input voltage. Since the inductor current has to be continuous, diode D_2 is forward biased and the capacitor C_3 is charged.

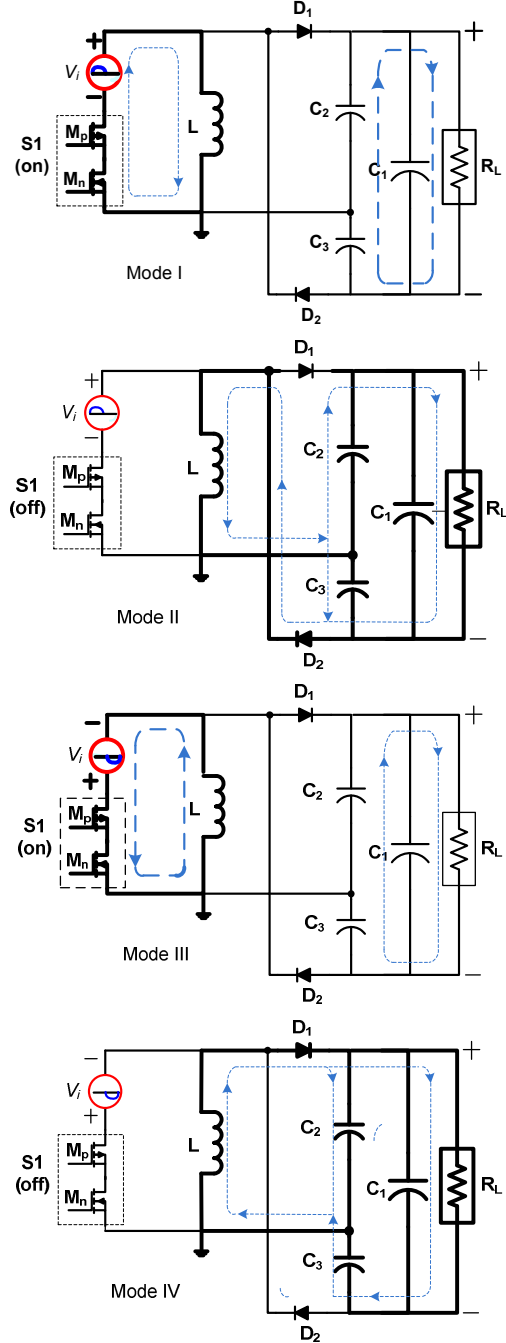


Fig. 4. Operating modes of the proposed Converter

2) Negative half-cycle:

Mode III: The switch S1 is turned on but the current builds

up in negative direction due to change in input voltage polarity.

Mode IV: When the switch is turned off, the inductor charges the capacitor C2 through diode D1. The capacitors C1, C2 and C3 share energy through charge recycling. Even though, the capacitors C2 and C3 show large variations in voltage; the duty cycle can be effectively controlled to maintain a steady DC voltage at output capacitor C1. It should be noted that the converter has an additional discontinuous mode in which the inductor current is zero.

B. Analysis

For analysis, the input is considered to have a peak voltage V_p with a time-period T_i . The converter operating with a switching time period, T_s , produces a steady output voltage V_0 with load resistance R_L for a duty cycle, D . Since, the converter is operating in discontinuous conduction mode; the input current over a line cycle can be represented as shown in Fig. 5. The inductor current is same as the input current when the switch is turned on. When the switch turns off, the input current becomes zero while the inductor current discharges to the output capacitors (Fig. 6). The switching cycle time period (T_s) is assumed to be much smaller than the line cycle (T_i). For a general k^{th} switching cycle, input voltage can be represented as:

$$V_{ik} = V_p \sin(2\pi k T_s / T_i) \quad (1)$$

The peak of the input current can be derived as:

$$i_{pk} = m_1 D T_s = V_{ik} D T_s / L \quad (2)$$

Since, the converter is operating in discontinuous conduction mode, the fall time ($d_f T_s$) can be estimated by inductor voltage balance on inductor L.

$$d_f T_s = i_{pk} / m_2 = i_{pk} L / (V_0 - V_{ik}) \quad (3)$$

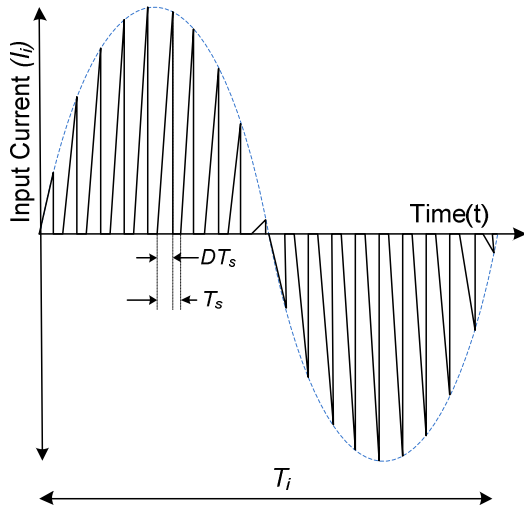


Fig. 5. Input Current over a line cycle

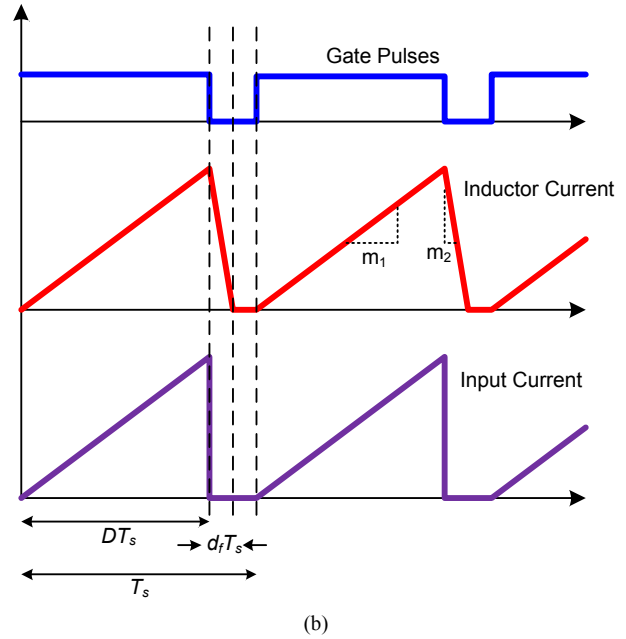


Fig. 6. Gate Pulses, Inductor Current, Input Current over a few switching cycles

The average power extracted from input over a switching cycle can be derived as:

$$P_{ik} = V_{ik} i_{pk} D / 2 \quad (4)$$

The total power over a line cycle can be derived by summation of this average power. Consider $N = T_i / T_s$ to be number of switching cycles in a line cycle. Therefore total input power can be written as:

$$P_i = (1/T_i) \sum_{k=1}^N V_{ik} i_{pk} D T_s / 2 \quad (5)$$

Since, N is a large number, the summation can be approximated as integration and the input power is derived as:

$$P_i = \frac{V_p^2 D^2 T_s}{4L} \quad (6)$$

where D is the duty cycle of the converter.

It can be seen that the extracted power only depends on the input characteristics and the duty cycle of the converter. This information can be used to design the passive components of the converter for a particular power level. The converter offers a resistive load, R_e to the microgenerator which can be derived as:

$$R_e = \frac{V_p^2}{2P_i} = \frac{2L}{D^2 T_s} \quad (7)$$

Therefore, it can be deduced that the converter offers an adjustable resistive load to the microgenerator [12]. Such a structure is very conducive to optimum energy harvesting.

III. SIMULATION RESULTS

The input is considered to be a sinusoidal input voltage with a frequency of 108 Hz. The simulation results are presented to demonstrate the capability of the converter to maintain a nominal DC voltage of 3.3 V. The load resistance is considered to be 1k Ω . The n-p MOSFET pair, Si4511DY from Vishay is utilized to form the bidirectional switch. The rectifying diodes D1 and D2 are schottkey diodes. The models for the devices are used with relevant parasitics. The converter is operated at a switching frequency of 50 kHz. The design of the inductor value is 10 μ H. The split capacitors (C2, C3) have a value of 10 μ F while the output capacitor (C1) has a value of 22 μ F.

To validate the boost operation of the converter, the input voltage is considered to have a peak of 250 mV. The input voltage and input current are shown in Fig. 7. It can be seen that the input current follows the input voltage as expected in DCM operation. The duty cycle required to reach 3.3 V is 0.8. The split capacitor voltages (V_{C2} and V_{C3}) are shown along with output voltage is Fig. 8. It can be seen that while the output voltage is fairly constant, the split capacitor voltages show large ripple as they are charged and discharged in alternate half-cycles. The current waveforms over a few switching cycles are shown in Fig. 9. When the gate signal goes low, the input current becomes zero while the inductor falls to zero in fall time $d_f T_s$ while charging the output capacitors. The estimated efficiency of the converter is around 65%.

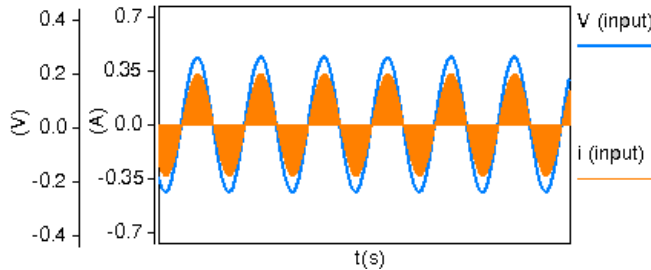


Fig. 7. Input Voltage (0.2V/div) & Input current (0.3A/div)

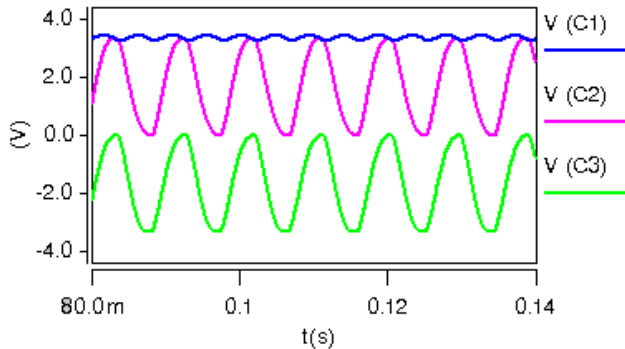


Fig. 8. Output Voltage, V_{C1} , Split Capacitor Voltages (V_{C2} & V_{C3}) (1V/div)

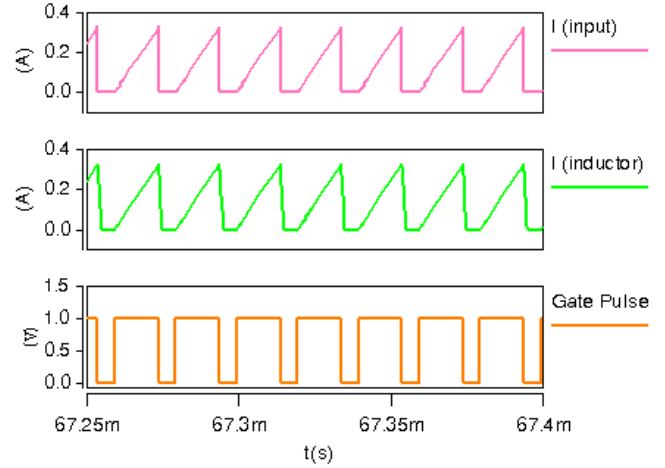


Fig. 9. Input Current (0.2A/div), Inductor Current (0.2A/div), Gate Pulses

As stated before, DCM operation is preferred, since it allows a simple voltage feedback control of the converter. A simple PI controller as shown in Fig. 10 is utilized to maintain the output voltage at 3.3V. The low pass filter is used to get rid of the second order line frequency ripple from the output voltage.

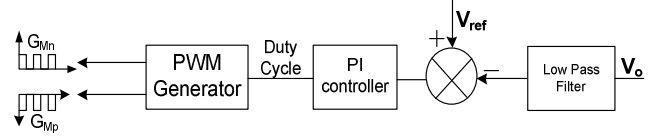


Fig. 10. Control Structure for the converter

The validity of the controller is verified by testing the ability to maintain constant output voltage in case of step change in load. The simulation results are presented in Fig. 11. At $t=0.4$ s, the load is changed from 1k Ω to 2k Ω . The controller changes the duty cycle from $D=0.75$ to $D=0.5$ while the output voltage remains constant at 3.3 V.

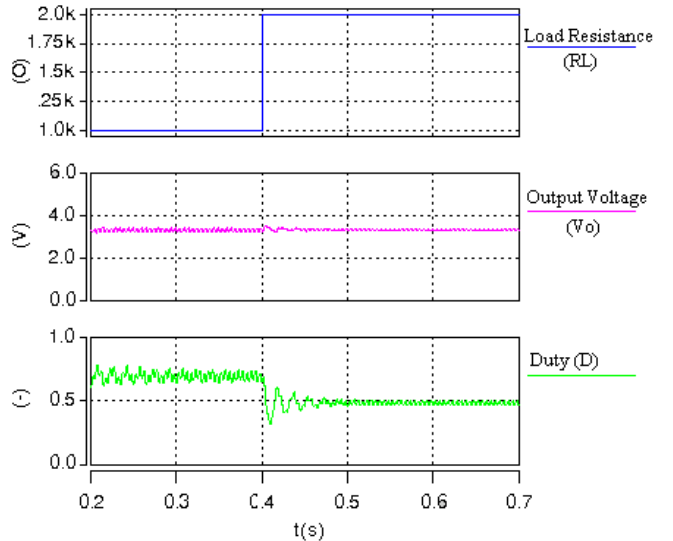


Fig. 11. From top to bottom trace : Load Resistance R_L (25k Ω /div), Output Voltage V_o (2V/div), Duty Cycle (.5 units/div)

IV. EXPERIMENTAL RESULTS

The prototype of the converter was fabricated. The relevant parasitic of the chosen converter components is shown in TABLE I. An electromagnetic microgenerator

TABLE I
CONVERTER COMPONENTS

Component	Name	Rating	Parasitic
Capacitor	C1	22 μ F	$R_{esr}=20\text{m}\Omega$
Capacitor	C2 & C3	10 μ F	$R_{esr}=50\text{m}\Omega$
Inductor	L	10 μ H	$R_{esr}=90\text{m}\Omega$
N - channel MOSFET	M_n	20V, 6A	$R_{ds_on} = 17\text{m}\Omega$ @ $V_g=3\text{ V}$
P-channel MOSFET	M_p	20V, 6A	$R_{ds_on} = 30\text{m}\Omega$ @ $V_g=3\text{ V}$
Schottky Diode	D1, D2	23V, 1A	$V_f=0.25\text{V}$ @15mA

capable of producing around 0.3 V AC at 108 Hz is utilized to verify the operation of the converter. The load resistance for the converter is 1 k Ω . The switching frequency of operation is 50 kHz. The converter is operated at a duty cycle, $D=0.8$ to reach a nominal DC voltage of 3.3 V. The current flowing in the inductor is shown in Fig. 12. It has a sinusoidal profile similar to simulation results. The inductor current over a few switching cycles is shown in Fig. 13. It can be seen that the converter operates in discontinuous conduction mode. The gate voltage for n-MOSFET (M_n) is positive when the p-MOSFET (M_p) gate voltage is negative; hence, they are switched on at the same time. The output voltage along with the voltage across capacitors C2 & C3 are shown in Fig. 14. Even though, the auxiliary capacitors, C2 and C3 show variation in voltage; the capacitor C1 has a steady output voltage. The efficiency of the converter is estimated to be around 61% which is quite close to the simulation results.

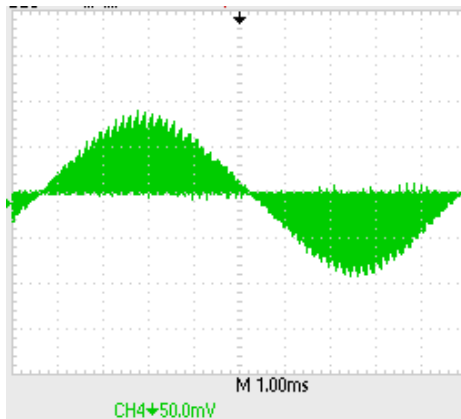


Fig. 12. Inductor Current (0.25A/div)

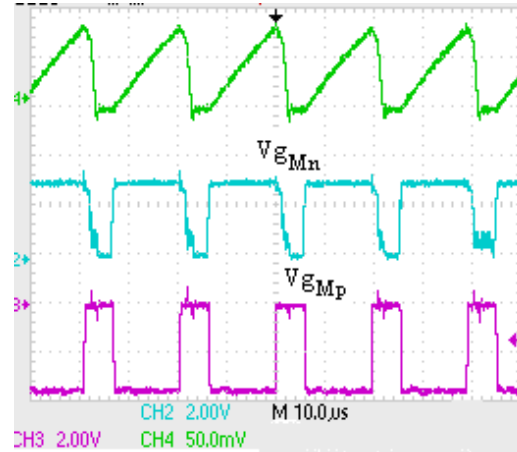


Fig. 13. Inductor current over a few switching cycles

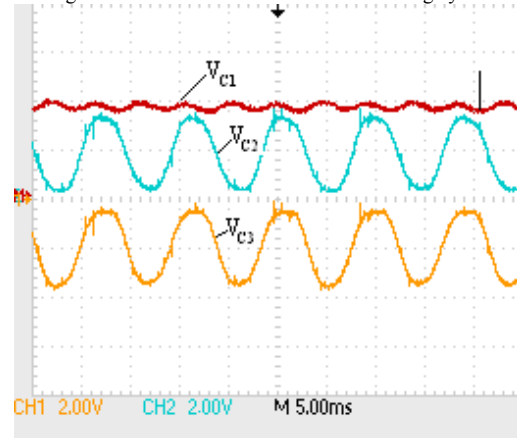


Fig. 14. Voltages across the capacitors C1, C2, C3 (2V/div).

V. CONCLUSION

In this paper, a new AC-DC converter is presented for low voltage; low power energy harvesting. The converter avoids front-end diode bridge rectification to achieve high efficiency. The buck-boost based configuration allows the duty cycle to completely control the harvested energy irrespective of the load characteristics. Simulation and experimental results demonstrate the ability of the converter to boost the low AC voltage to a steady DC voltage.

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